BenchFlow — Home page

BenchFlow

Prêt hors ligne

Welcome to BenchFlow!

Your experiments start here.

- Home page
- Procol Library
- Smart Experimental Workflow
- Scientific Calculator

Logo of BenchFlow

Slogan de BenchFlow

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BenchFlow

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Protocol Library

Select a protocol to view its steps and start a new experimental session.

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DNA Extraction

6 steps

View Protocol →

Solution Dilution

4 steps

View Protocol →

Colorimetric Assay

5 steps

View Protocol →

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Lecture du protocole

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Smart Experimental Workflow

Choose or start a new session

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Saved Sessions

+ New Session

3 sessions

DNA Extraction — In progress
EXP-001 · Step 3/6 — 50%
Resume Session →

Solution Dilution — Paused
EXP-002 · Step 2/4 — 50%
Resume Session →

Colorimetric Assay — Completed
EXP-003 · Step 5/5 — 100%
View Session →

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DNA Extraction

Extraction and purification of genomic DNA from a biological sample.

6 steps

~ 30–45 min

Objective

Extract and purify genomic DNA from a biological sample (e.g. cells, tissues or blood) for downstream analyses.

Required Materials

- DNA extraction kit
- 1.5 mL microtubes
- Centrifuge
- Micropipettes and tips
- Wash solution
- Nuclease-free water

Expected Results

Purified DNA of sufficient quality for downstream applications such as PCR, qPCR or sequencing.

Protocol Steps

1. **Cell Lysis**
   Add the lysis buffer and homogenize the sample.
2. **Protein Precipitation**
   Add the precipitation buffer and centrifuge.
3. **Supernatant Recovery**
   Transfer the supernatant containing the DNA to a new tube.
4. **DNA Precipitation**
   Add isopropanol and incubate.
5. **DNA Washing**
   Wash the pellet with 70% ethanol.
6. **DNA Elution**
   Resuspend the DNA in nuclease-free water.

▶ Start Experiment

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DNA Extraction > Session EXP-001

1 Cell Lysis

Add the lysis buffer and homogenize the sample.

~ 10 min | Materials (5 items) | Notes allowed

Required Materials

- 1.5 mL microtubes
- Lysis buffer
- Micropipette (P200)
- Sterile pipette tips
- Vortex mixer

Experiment Notes

Record your observations, volumes, remarks...

B I U

Instructions

1. Prepare the sample in a 1.5 mL microtube.
2. Add 200 µL of lysis buffer.
3. Homogenize using a vortex mixer for 30 seconds.
4. Incubate at room temperature for 10 minutes.

Previous Step | Save and Exit | Next Step

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Scientific Calculator

Tools for your laboratory calculations and preparations.

Dilution Calculator

Easily calculate the volumes to pipette and to complete using the formula $C_1V_1 = C_2V_2$.

$$C_1 \times V_1 = C_2 \times V_2$$

C_1 : stock solution concentration
V_1 : volume of stock solution to pipette
C_2 : desired final concentration
V_2 : final volume of solution

Stock concentration (C_1): 1 mg/mL

Final concentration (C_2): 0.1 mg/mL

Final volume (V_2): 10 mL

Volume to pipette (V_1): 1 mL

Calculate | Reset

Result

Pipette **1 mL** of the stock solution (C_1 = 1 mg/mL) and complete with solvent up to **10 mL** (C_2 = 0.1 mg/mL).

Timer

Start a timer to track the duration of a protocol step.

05:00
min sec

Start | Pause | Reset

Preset duration

1 min | 5 min | 10 min | 30 min

Volume Converter

Convert between µL and mL.

500 µL ↔ 0.5 mL

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